Computational Hydrology and Remote Sensing

The *Computational Hydrology and Remote Sensing* research group at Rice University focuses on the intersection between water resources, food, and climate. Our research aims to support actionable decision-making by improving hydrological information for monitoring and forecasting of hydrological extremes and their impacts at local scales. To this end, we develop scalable computational approaches for hyper-resolution hydrological prediction targeting regional to global-scale applications by leveraging advances in satellite remote sensing, land surface modeling, artificial intelligence, big geospatial data, data fusion, and high-performance computing. More information at www.waterai.earth.

Our research group works on multidisciplinary projects in the following areas (and at their intersections) :

1. Advancing terrestrial hydrology through physics-based AI
2. Satellite remote sensing and AI data assimilation
3. Integrated Earth system modeling and climate change impacts
4. Prediction of extreme hydroclimate events (floods, droughts, landslides, wildfires)
5. Quantitative assessments of climate change and human interventions on water scarcity

Positions Open:

1. PhD – Hydrologic Preconditions for Landslides (Spring 2027)
Develop a high-resolution physics-informed landslide-risk model that accounts for hydrological preconditions by integrating AI, hydrological modeling, and satellite data fusion. This project will support the USDA Forest Service's CulvertApp (https://culvert-at-risk.org/) in partnership with the Department of Transportation.
Application deadline: **October 30, 2026.**

2. PhD – Drought Modeling and Prediction (Fall 2027)
This NSF-funded project will combine land surface modeling, satellite observations, and farmer-reported drought data to identify local drought typologies and improve drought detection and prediction.
Application deadline: **January 8, 2027.**

3. Postdoctoral Researcher — Graph AI for Geoscience (Spring 2027)
As part of our Multimodal Geosensing AI for Earth System Discovery research cluster funded by the Rice University Ken Kennedy Institute, the postdoctoral researcher will lead the development of foundational graph neural networks and multimodal AI methods for complex geospatial systems. More information here. Application deadline: **December 1st, 2026.**

4. Pan Postdoc Fellowship (Spring or Fall 2027)
The Pan Fellowship supports early-career researchers in pursuing original, independent research in collaboration with one or more Rice EEPS faculty members. Applicants interested in computational hydrology, remote sensing, geospatial AI, or related topics are encouraged to reach out before applying.
Application deadline: **October 25, 2026.**

Background:

Successful candidates will have a background in geosciences, environmental science and engineering, climate sciences, applied math, physics, scientific deep learning, or related fields. This is an excellent

opportunity to develop expertise in land surface modeling, hydrologic prediction, satellite remote sensing, and data science.

Essential Qualifications:

- Strong analytical skills, ability to think critically and solve problems effectively.
- A solid understanding of calculus and numerical methods.
- Strong programming skills (UNIX/Linux, Python, shell scripting) for processing and visualization of simulation and remote sensing data.
- Experience with geospatial datasets (e.g., GeoTIFF, NetCDF, HDF5, Zar, dask) and data processing using Python geospatial libraries (e.g., xarray, cartopy, rasterio).
- Experience with machine learning and/or deep learning concepts using Python (e.g., PyTorch, TensorFlow)
- Eagerness to acquire new skills and adapt them to cutting-edge research in hydrology and Earth sciences.
- Excellent written and verbal communication.

Desired Qualifications:

- Demonstrate a research track record of involvement in topics relevant to computational hydrology, remote sensing, or related fields.
- Experience with at least one advanced programming language (e.g., C, C++, Fortran, etc.)
- Hands-on experience with machine learning or deep learning frameworks (e.g., TensorFlow, PyTorch).
- Familiarity with High-performance computing (HPC) systems and/or cloud computing (e.g., AWS, Google Cloud, etc.).
- Familiarity with version control systems for collaborative code development (e.g., Github, Gitlab)

All researchers will benefit from our group's involvement in national and international collaborative projects, and Rice's thriving and expanding programs, such as the Rice Data Science Initiative, Data to Knowledge Lab, and the Ken Kennedy Institute.

Application for PhD students:

PhD students should submit an application to Earth, Environmental, and Planetary Sciences (deadline of **October 30th 2026 for the Spring 2027** and **January 8th 2027 for the Fall 2027**). International students should also meet the language proficiency requirements. Prospective graduate students can email Dr. Vergopolan (vergopolan@rice.edu) with the subject "Prospective PhD student" before applying. In the email, please include your unofficial transcripts and curriculum vitae attached as a PDF and include *a brief personal statement explaining why you fit and why you would like to join the group*. We greatly appreciate all the applications, but due to the high volume of submissions, only shortlisted candidates will be notified. Compensation: $37K/year stipend with benefits plus full tuition ($66K/year).

Postdoctoral Researchers:

Outstanding applicants may explore fellowship opportunities, including the Pan Postdoctoral Research Fellowship (deadline on **October 25th, 2026**), the UCAR Climate & Global Change Postdoctoral Program (deadline on **October 16th, 2026**), among others. Applicants should have demonstrated expertise in process-based modeling, remote sensing, or land surface modeling, along with strong programming skills and experience in HPC or cloud computing. Excellent written and verbal communication, demonstrated through a strong publication record is expected. Candidates can reach out and send their CV, transcripts, a brief statement explaining your interest, and contact information of 3 references to Dr. Vergopolan (vergopolan@rice.edu) with the subject line “Prospective Postdoctoral Researcher.” We

greatly appreciate all the applications, but due to the high volume of submissions, only shortlisted candidates will be notified. Salary is commensurate with experience, ranging from $60k–70k per year, plus benefits.